Written Homework 2. 20210356 Chihyun Ahn.

1. $Q_n(S_t, A_t) = \frac{1}{n}\sum_{i=1}^{n} G_i(S_t, A_t)$ and $Q_{n-1}(S_t, A_t) = \frac{1}{n-1}\sum_{i=1}^{n-1} G_i(S_t, A_t)$

$$\therefore Q_n(S_t,A_t) = \frac{1}{n}\sum_{i=1}^{n} G_i(S_t,A_t) = \frac{1}{n}\left(\sum_{i=1}^{n-1} G_i(S_t,A_t) + G_n(S_t,A_t)\right)$$

$$= \frac{n-1}{n}\left(\frac{1}{n-1}\sum_{i=1}^{n-1} G_i(S_t,A_t)\right) + \frac{1}{n}G_n(S_t,A_t)$$

$$= \frac{1}{n-1}\sum_{i=1}^{n-1} G_i(S_t,A_t) - \frac{1}{n}\left(\sum_{i=1}^{n-1} G_i(S_t,A_t)\right) + \frac{1}{n}G_n(S_t,A_t)$$

$$= Q_{n-1}(S_t,A_t) + \frac{1}{n}\left(G_n(S_t,A_t) - Q_{n-1}(S_t,A_t)\right)$$

2. The observed episode should lasts 10 steps. $\therefore$ Agent stay at A for 9 times and reach B at 10th step. $\therefore$ A → A → A → A → A → A → A → A → A → A → B

p

↳ 1-p.

⇒ This is one episode

$\therefore$ (1) With the first Visit Monte Carlo,

Value of the state A is $\frac{1}{1} \times 10 = 10.$

(2) With every visit Monte Carlo,

Value of the state A is $\frac{1}{10}(10+9+8+\cdots+2+1) = \frac{55}{10} = 5.5$

3. $V(S_0) = V(S_0) + \alpha\left(Reward_\pi + \gamma V_\pi(S_{right}) - V(S_0)\right)$

$$= V(S_0) + \alpha\left(\frac{\pi(a_1|S_0)}{\mu(a_1|S_0)}\left(Reward_\mu + \gamma V_\mu(S_{right})\right) - V(S_0)\right)$$

$$= 5 + \frac{1}{10}\left(4\left(-1 + \frac{9}{10}\times 6\right) - 5\right) = 5 + \frac{1}{10}(17.6 - 5) = 5 + \frac{1}{10}\times 12.6$$

$= 6.26$

4. $V(S) = V(S) + \alpha\left(G_\pi(S) - V(S)\right),$

$G_\pi(S) = 4\times 4\times 4\times 4\times G_\mu(S).$

$G_\mu(S) = -1 + 0.9\times(-1) + (0.9)^2\times(-1) + (0.9)^3\times 10$

$= -1 - 0.9 - 0.81 + 7.29 = 4.58$

$G_\pi(S) = 1172.48$

$\therefore V(S) = 5 + \frac{1}{10}(1172.48 - 5) = 5 + 116.748 = 121.748$

5. NO. Q-learning update value function through

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha(R_t + r\max(Q(s_{t+1}, a)) - Q(s_t, a_t))$$

And SARSA update value function through

$$Q(s_t, a_t') = Q(s_t, a_t') + \alpha(R_t + rQ(s_{t+1}, a_{t+1}') - Q(s_t, a_t'))$$

and $a_{t+1}' = \mathrm{argmax}_a Q(s_{t+1}, a)$

$\therefore Q(s_{t+1}, a_{t+1}') = \max(Q(s_{t+1}, a))$

It looks like same but the way of choosing a_t and a_t' is different. for SARSA, $a_t' = \mathrm{argmax}_a Q(s_t, a)$

But Q-learning $a_t = \begin{cases} \mathrm{argmax}_a Q(s_t, a) & \text{for } 1-\varepsilon \\ \text{random } a & \text{for } \varepsilon \end{cases}$

$\therefore$ Q-learning can explore where the state is not greedy.

$\therefore$ Different.

For example. If the initial value of $Q(s_t, a_1) = Q(s_t, a_2) = 0$.

And the reward is $R_{st}^{a_1} = +1$, $R_{st}^{a_2} = +2$,

With SARSA It only select a_1 and can not select a_2 any more in s_t.

However, Q-learning can select a_2 sometimes and can reach optimal.

6. (1) $r^n = r_t r_{t+1} r_{t+2} \cdots r_{t+n} = \prod_{i=0}^{n} r(s_{t+i})$, $r_t = r(s_t) = 1$.

$\therefore r^n = \prod_{i=1}^{n} r(s_{t+i})$

$\therefore G_t = \sum_{n=0}^{\infty} r^n R_{t+n+1} = \sum_{n=0}^{\infty} \prod_{i=1}^{n} r(s_{t+i}) R_{t+n+1}$

(2) $G_{t+1} = \sum_{n=0}^{\infty} \prod_{i=1}^{n} r(s_{t+1+i}) R_{t+n+2}$,

$$G_t = \sum_{n=0}^{\infty} \prod_{i=1}^{n} r(s_{t+i}) R_{t+n+1}$$

$$= R_{t+1} + \sum_{n=1}^{\infty} \prod_{i=1}^{n} r(s_{t+i}) R_{t+n+1}, \quad (\text{let } n-1 = k)$$

$$= R_{t+1} + \sum_{k=0}^{\infty} \prod_{i=1}^{k+1} r(s_{t+i}) R_{t+k+2} = R_{t+1} + \sum_{k=0}^{\infty} \prod_{p=0}^{k} r(s_{t+p+1}) R_{t+k+2} \quad (p = i-1)$$

$$= R_{t+1} + \gamma(S_{t+1}) \sum_{k=0}^{\infty} \prod_{p=1}^{k} \gamma(S_{t+1+p}) R_{t+k+2}$$

$$= R_{t+1} + \gamma_{t+1} G_{t+1}$$

$$\therefore G_t = R_{t+1} + \gamma_{t+1} G_{t+1}$$

(3) $V(S_t) = V(S_t) + \alpha(R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$

$$= V(S_t) + \alpha(R_{t+1} + \gamma(S_{t+1}) V(S_{t+1}) - V(S_t))$$

$$= V(S_t) + \alpha(R_{t+1} + \gamma_{t+1} V(S_{t+1}) - V(S_t))$$